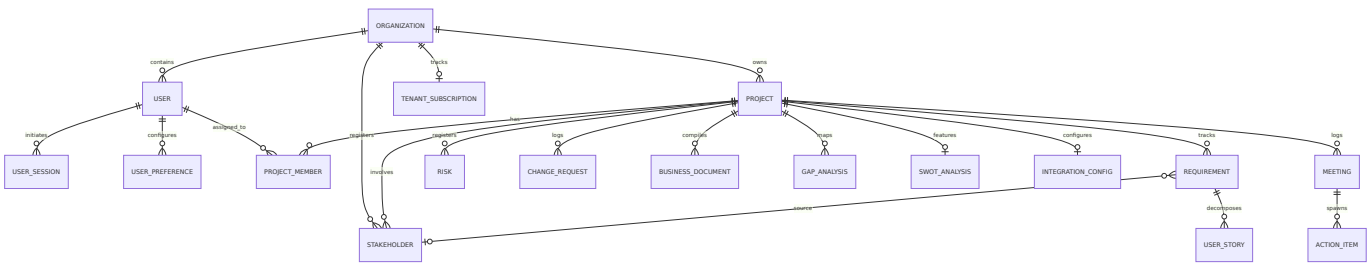


BAHub - Phase 6: Data Analysis & Database Schema Mapping

This document details the database architecture of **BAHub**, mapping entity relationships, structural field tables, foreign key constraints, and classifying application data.

1. Entity Relationship Diagram (ERD)

The diagram below represents the logical database schema of BAHub. All entities inherit audit tracking properties (`id` UUID primary key, timestamps, `is_deleted` soft-delete flag) from the abstract `BaseModel` .



2. Master Data vs. Transaction Data

To optimize system performance and understand access patterns, BAHub classifies data into two core types:

2.1 Master Data (Core Entities)

This data is relatively static and defines the foundational building blocks of the workspace:

- **Organizations:** Tenant container accounts.
- **Users:** Identity profiles and system role definitions.
- **Projects:** Scoped collaboration spaces.
- **Stakeholders:** Directory of names, titles, and departments.
- **Integration Configs:** Connection details for third-party systems like Jira and Confluence.

2.2 Transaction Data (Operational Logs)

This data changes frequently, generated by users' daily operational actions:

- **Requirements & User Stories:** Logged specifications and Agile cards.
- **Meetings & Action Items:** Minutes of meetings and checklist items.
- **Risks & Change Requests:** Audits and approval logs.
- **SWOT & Gap Analyses:** Business analysis worksheets.
- **User Sessions:** Login logs.

3. Data Dictionary & Table Definitions

3.1 Table: `organizations` (Organization Model)

- **Description:** The top-level multi-tenant boundary. All project data is isolated within an organization context.

- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Unique organization identifier.
name	VARCHAR(255)	Not Null	Unique	-	Name of the organization.
logo	VARCHAR(100)	Nullable	-	-	Path to the uploaded logo file.
description	TEXT	Nullable	-	""	Optional description of the organization.
timezone	VARCHAR(100)	Not Null	-	"UTC"	Active timezone for organization operations.
email	VARCHAR(254)	Nullable	-	-	General contact email.
phone	VARCHAR(50)	Nullable	-	-	Contact phone number.
website	VARCHAR(200)	Nullable	-	-	URL of the organization website.
address	TEXT	Nullable	-	""	Physical address.
created_at	DATETIME	Not Null	-	Auto Now Add	Creation timestamp.
updated_at	DATETIME	Not Null	-	Auto Now	Last update timestamp.
is_deleted	BOOLEAN	Not Null	-	False	Soft delete flag.

3.2 Table: users (Custom User Model)

- **Description:** Represents workspace accounts, linking authentication details, roles, and organization scopes.

- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Unique user identifier.
username	VARCHAR(150)	Not Null	Unique	-	Login username.
password	VARCHAR(128)	Not Null	-	-	Hashed password.
email	VARCHAR(254)	Not Null	-	-	User email.
role	VARCHAR(50)	Not Null	-	"BUSINESS_ANALYST"	Choices: ADMIN , BUSINESS_ANALYST , PRODUCT_OWNER , DEVELOPER , QA_TESTER , STAKEHOLDER .
organization_id	UUID	Nullable	Foreign Key -> organizations(id)	SET_NULL	Links user to their organization.

phone	VARCHAR(30)	Nullable	-	-	User phone number.
bio	TEXT	Nullable	-	-	User profile description.
is_active	BOOLEAN	Not Null	-	True	Django status flag.
is_staff	BOOLEAN	Not Null	-	False	Django admin access flag.

3.3 Table: projects (Project Model)

- **Description:** Scoped collaboration spaces within an organization.
- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Unique project identifier.
organization_id	UUID	Not Null	Foreign Key -> organizations(id)	CASCADE	Owner organization context.
name	VARCHAR(255)	Not Null	Unique with organization_id	-	Project name.
description	TEXT	Nullable	-	" "	Project description.
status	VARCHAR(50)	Not Null	-	"ACTIVE"	Choices: ACTIVE , COMPLETED , ARCHIVED .
start_date	DATE	Nullable	-	-	Planned start date.
end_date	DATE	Nullable	-	-	Target completion date.

3.4 Table: project_members (ProjectMember Model)

- **Description:** Junction table mapping users to projects and defining project-specific roles.
- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Junction ID.
project_id	UUID	Not Null	Foreign Key -> projects(id)	CASCADE	Linked project.
user_id	UUID	Not Null	Foreign Key -> users(id)	CASCADE	Assigned user.
role	VARCHAR(50)	Not Null	Unique with project_id , user_id	"CONTRIBUTOR"	Project role. Choices: PROJECT_MANAGER , CONTRIBUTOR , VIEWER .

3.5 Table: stakeholders (Stakeholder Model)

- **Description:** Registry of stakeholders involved in requirements discovery.

- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Stakeholder ID.
organization_id	UUID	Not Null	Foreign Key -> organizations(id)	CASCADE	Organization context.
project_id	UUID	Nullable	Foreign Key -> projects(id)	CASCADE	Optional project scoping.
name	VARCHAR(255)	Not Null	-	-	Full name of stakeholder.
title	VARCHAR(255)	Not Null	-	-	Job title (e.g. CPO, Director).
department	VARCHAR(255)	Nullable	-	" "	Stakeholder department.
email	VARCHAR(254)	Nullable	-	-	Contact email.
power	VARCHAR(50)	Not Null	-	"LOW"	Choices: HIGH , LOW .
interest	VARCHAR(50)	Not Null	-	"LOW"	Choices: HIGH , LOW .
influence	INTEGER	Not Null	-	3	Score from 1 to 5.
impact	INTEGER	Not Null	-	3	Score from 1 to 5.

3.6 Table: requirements (Requirement Model)

- **Description:** Detailed catalog of project specifications.

- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Requirement ID.
project_id	UUID	Not Null	Foreign Key -> projects(id)	CASCADE	Linked project.
req_id	VARCHAR(50)	Not Null	Unique with project_id	Auto-calculated	Sequential identifier (e.g. REQ-001).
title	VARCHAR(255)	Not Null	-	-	Brief requirement title.
description	TEXT	Not Null	-	-	Detailed specification text.
req_type	VARCHAR(50)	Not Null	-	"FUNCTIONAL"	Choices: FUNCTIONAL , NON_FUNCTIONAL , TECHNICAL , UI .

status	VARCHAR(50)	Not Null	-	"DRAFT"	Choices: DRAFT , REVIEW , APPROVED , REJECTED .
priority	VARCHAR(50)	Not Null	-	"MEDIUM"	Choices: HIGH , MEDIUM , LOW .
version	VARCHAR(20)	Not Null	-	"1.0"	Document version tracker.
source_stakeholder_id	UUID	Nullable	Foreign Key -> stakeholders(id)	SET_NULL	Tracing source stakeholder.

3.7 Table: user_stories (UserStory Model)

- **Description:** Decomposed agile stories linked to parent requirements.
- **Fields:**

Field Name	Data Type	Nullability	Constraints / Keys	Default Value	Description
id	UUID	Not Null	Primary Key	uuid4()	Story ID.
requirement_id	UUID	Not Null	Foreign Key -> requirements(id)	CASCADE	Parent requirement.
story_id	VARCHAR(50)	Not Null	Unique with requirement_id	Auto-calculated	Sequential story ID (e.g. US-001).
title	VARCHAR(255)	Not Null	-	-	Story title.
role	VARCHAR(255)	Not Null	-	-	"As a..." actor.
action	TEXT	Not Null	-	-	"I want to..." action description.
benefit	TEXT	Not Null	-	-	"So that..." benefit statement.
acceptance_criteria	TEXT	Nullable	-	" "	Gherkin format validation scripts.
status	VARCHAR(50)	Not Null	-	"TODO"	Choices: TODO , IN_PROGRESS , QA , DONE .
points	INTEGER	Not Null	-	3	Fibonacci scale: 1, 2, 3, 5, 8, 13.
jira_key	VARCHAR(100)	Nullable	-	-	External JIRA key mapping (e.g. PROJ-12).

jira_url	VARCHAR(512)	Nullable	-	-	URL link to synchronized JIRA issue.
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